

Report on the experimental project for the course Statistical Methods for Machine Learning

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Abstract. This report is focused on the assignment regarding online learning and in particular the adaptive hedge algorithm. After an introduction where the problem has been outlined and an analysis on the classical fixed hedge algorithm and on his variant adaptive hedge, there is a discussion where the methodologies of the experiment are presented and the results are explained.

1 Introduction: Continuous Prediction with Experts Advice

The continuous prediction with expert advice is one of the fundamental problems of online learning, it can be presented as follows:

An agent has across time the possibility to make certain choices regarding a problem, in particular during a sequence of turns $t_1, t_2, \dots$ the agent has to decide in each turn which action is the best available for him, in doing so he can rely on the advice of a pool of K experts each with unknown a priori reliability.

The action choice is immaterial: it could be interpreted as a move in a game of chess or as a policy choice for a president in charge, the only constant is that time is considered discrete and the fact that the result of the action can be viewed only on the successive turn.

So our agent, that doesn't understand anything on the actions he is performing, relies only on the advice of these experts each suggesting something, but having as only mean of evaluation their past performances.

How good of a choice the action was, can be viewed on the turn after by computing the loss function on the expert's advised action, so some expert might have given good advice $l_t(A_t) = 0$ while others not so much $l_t(B_t) = 1$, where A_t and B_t are well and bad performing actions on turn t , this of course can only be discovered at turn $t + 1$.

2 The Hedge Algorithm

Given the previous description of the problem the main characteristics are:

- A possible antagonistic environment
- The complete reliance on the advice of the K experts

- No assumption of any kind on the quality of the predictions

A first intuitive approach to the problem would be to design an algorithm that keeps "betting" on the prediction of the expert that so far has made the least amount of errors. This approach is called follow the leader, it performs fairly well in a stochastic environment, but in a worst case scenario can perform very poorly: imagine a situation with just 2 experts each being right alternatively every turn, this approach switch once a turn the expert to take advice from, getting almost the worst possible result while both the experts have a one out of two success rate.

The hedge algorithm offers a solution to this problem: instead of always choosing the best expert so far, each expert has a *weight* that changes each turn, the expert are drawn with probability equal to their weight over the total weight of all the experts. The weight is then updated each turn according to the quality of the prediction made on the previous turn.

Algorithm 1 Hedge Algorithm

Require: $\eta \in (0, 1)$
 $w_1(i) = 1 \forall i \in [1, K]$
for $t = 1, \dots, T$ **do**
 $p_t(i) = w_t(i)/W_t$ $\triangleright$ Where W_t is the sum of $w_t(i)$ for all i
 Draw I_t according to $p_t(i)$
 Incur in loss $l_t(I_t)$ reveal all losses $l_t(i)$
 $w_{t+1}(i) = w_t(i)e^{-\eta l_t(i)}$ $\triangleright$ Weights update
end for

By tuning the learning rate, the rate at which good or bad predictions modify the weights of the experts, $\eta = \sqrt{2\ln(K)/T}$ the main property of this algorithm is:

$$E \left[\frac{1}{T} \sum_{t=1}^T l_t(I_t) \right] - \min_{i=1,2,\dots,K} \left(\frac{1}{T} \sum_{t=1}^T l_t(i) \right) \leq \sqrt{\frac{2\ln(K)}{T}} \quad (1)$$

establishing an upper bound, in function of T and K, between the difference of the expected average of the sequential risk and the average loss of the best performing expert.

When $T \rightarrow \infty$ the difference between the two tends to 0 meaning that the sequential risk of Hedge converges to the average loss of the optimal expert, so the algorithm will perform (for $T \rightarrow \infty$) as well as the best expert.

3 Adaptive Hedge Algorithm

A limit of the hedge algorithm is that the learning rate is static, once fixed at the beginning of the computation, it remains the same. The general strategy is to choose η in order to optimize worst case performance, but at the same time this could mean sub-optimal performance in the easy instances.

A proposed solution is the variant of the algorithm call Adaptive Hedge:

Algorithm 2 Adaptive Hedge Algorithm(ϕ)

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 $\eta = \phi$ 
for  $t = 1, \dots, T$  do
  if  $t = 1$  or  $\Delta \geq b$  then
     $\eta = \eta / \phi$ 
     $b = \left( \frac{1}{e-1} + \frac{1}{\eta} \right) \ln(K)$ 
     $\Delta = 0$ 
     $(w^1, \dots, w^K) = (\frac{1}{K}, \dots, \frac{1}{K})$ 
  end if
  Output probabilities  $w$  for round  $t$ 
  Actions receive losses  $l_t$ 
   $\Delta = \Delta + \mathbf{w} l_t + \frac{1}{\eta} \ln(\mathbf{w} e^{-\eta l_t})$ 
   $\mathbf{w} = (w^1 \cdot e^{-\eta l_t^1}, \dots, w^K \cdot e^{-\eta l_t^K}) / (\mathbf{w} \cdot e^{-\eta l_t})$ 
end for

```

This algorithm varies the learning rate adapting to the difficulty of the problem, obtaining constant regret for easy instances while maintaining (1) for the worst case.

It works by creating new segments each time the algorithm exceeds the budget, the new segment is characterized by a different learning rate of value $\eta_i = \phi^{1-i}$ where i is the number of the segment.

It maintains a learning rate $\eta \geq \sqrt{(e-1)\ln(K)/L_T^*}$, still not that large because a bound of order $O(\sqrt{L_T^* \ln(K)})$ can still be provided for the worst case regret. It's possible to bound the regret in terms of the number of segments m generated by the algorithm (changes in learning rate), and in terms of the number of turns T :

$$R_{AdaHedge}(T) \leq 2\ln(K) \left(\frac{\phi^m - 1}{\phi - 1} \right) + m \left(\frac{1}{e-1} \ln(K) + \frac{1}{8} \right) \quad (2)$$

$$R_{AdaHedge}(T) \leq \frac{\phi\sqrt{\phi^2-1}}{\phi-1} \sqrt{\frac{4}{e-1} L_T^* \ln(K)} + O(\ln(L_T^* + 2)\ln(K))$$

(3)

Also in a probabilistic setting(not a worst case scenario), is possible to prove that regret is bounded as follows¹:

$$R_{AdaHedge}(T) = O(K + \log(1/\delta))$$

(4)

4 Set Up of the Experimental Project

The work started by developing a python program to generate synthetic losses called *datasetgeneration.py*, it allows to generate in a file a sequence of losses $(0, 1)$ organized in a matrix of any dimension, where the number of columns corresponds to K , the number of experts, and the number of rows to T , the number of turns, the binomial distribution generator has been utilized to generate such losses so that they remain independent from each other.

The algorithms for *hedge* and *adahedge* have been implemented completely from scratch, following the pseudo-code and utilizing only the *numpy* library to ease mathematical operations. Also each algorithm can be run consecutively a number n of times, for the experiments performed n was always 10.

Finally the visualization is handled through the *matplotlib* library and by recording, during the computation, the regret and cumulative losses at each turn, which are averaged at the end.

The hedge algorithm has been run using different η while adaptive hedge has been used with a fixed $\phi = 2$.

5 Results Analysis

All the tests are run with a $T=10000$, the graphs unit of measure for time is scaled by 100.

5.1 First Test

The first test performed was a control test, utilizing the same set up present in in the Ada Hedge paper¹ with four experts with a probability of incurring in losses of 0.3, 0.4, 0.45 and 0.5 respectively. The learning rate given to the hedge algorithm was $\eta = \sqrt{2\ln(K)/L_T^*}$ around $\eta = 0.03$, of course to calculate the learning rate the agent used post hoc knowledge. Both the algorithms were run 10 times and the means of their regret were plotted.

Adaptive hedge performed much better with an avarege regret of 9.9 and an $m=2$, while fixed hedge had a regret of 48.7.

From the data is possible to see that in this easy instance the learning rate for fixed hedge(obtained through the bound) is far too small, while the adaptive hedge algorithm can easily scale his learning rate accordingly.

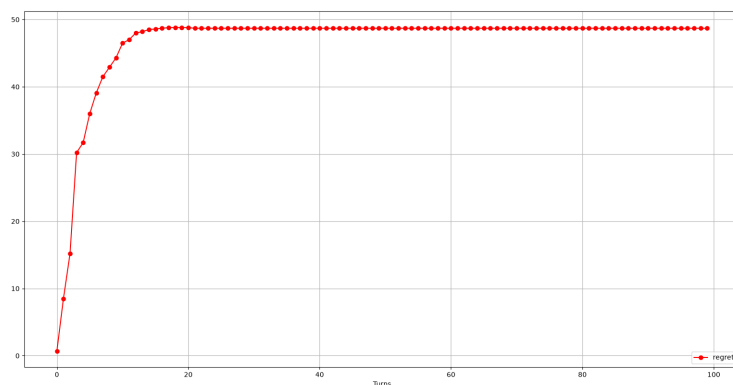
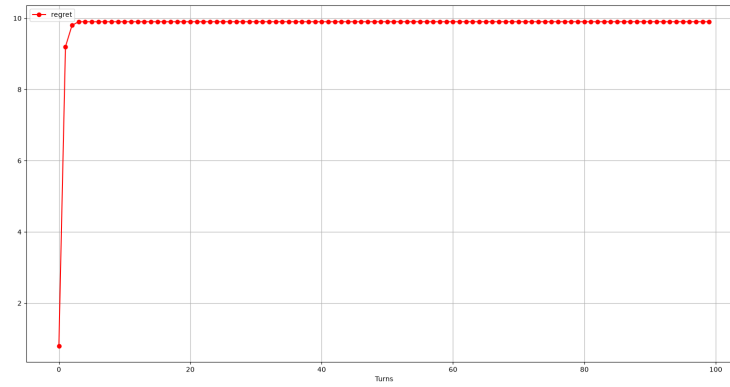
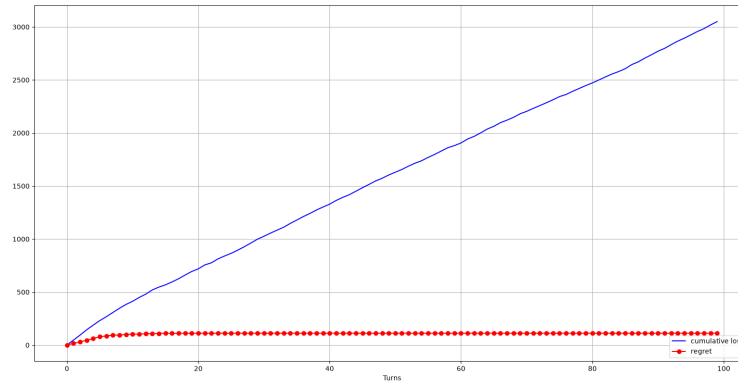


Fig. 1. Fixed Hedge

**Fig. 2.** Adaptive Hedge

5.2 Stochastic losses

The second experiment was on stochastic losses with 10 experts, one with probability 0.3 the other 9 with 0.5. Again this is a simple set up where one expert is significantly better than all the others, so expect adaptive hedge to perform much better. The learning rate for fixed hedge was set as $\eta = 0.02$, according to the bound in (1).

**Fig. 3.** Fixed Hedge average regret=112

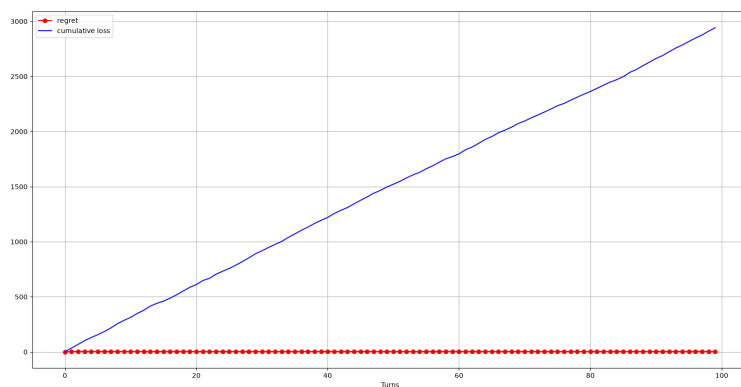


Fig. 4. Adaptive Hedge average regret=3.8

5.3 Adversarial losses

In this case 100 experts are used, once every turn an expert get a probability of 0.3 instead of 0.5. Still it wasn't enough, again with 10000 turns Ada Edge performs far better, then fixed hedge with $\eta = 0.03$ again obtained through (1).

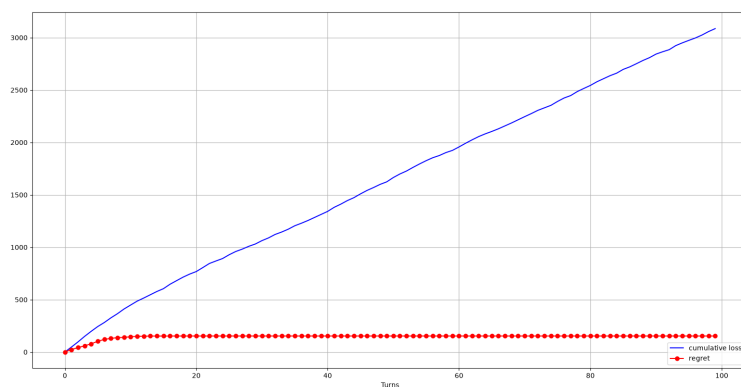


Fig. 5. Fixed Hedge has an average regret=153.1

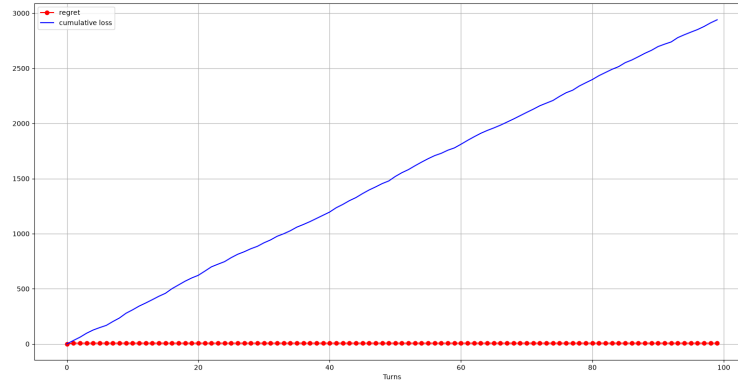


Fig. 6. Adaptive Hedge has an average regret=7.2

5.4 Low-Gap Losses

In this case, a worst case scenario for adaptive hedge is created: 99 experts with 0.5 as probability and one expert with 0.49, so there isn't a significant exploitable structure to get from the data, also the learning rate for fixed hedge was set at $\eta = 0.06$ according to $\eta = \sqrt{8\ln(K)/T}$. The results are slightly better for fixed edge but adaptive hedge still maintains a significant advantage in terms of regret.

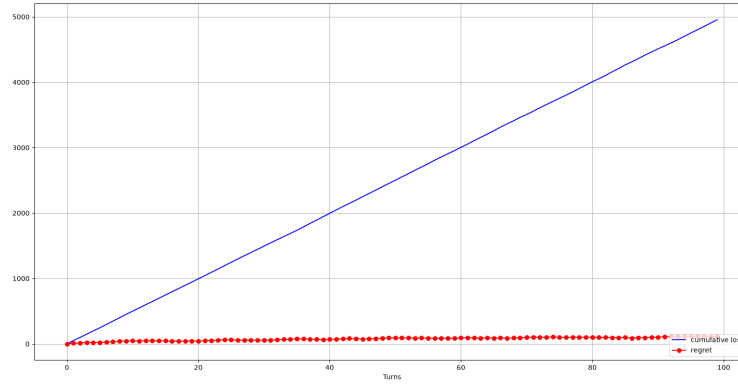


Fig. 7. Fixed Hedge average regret=116.1

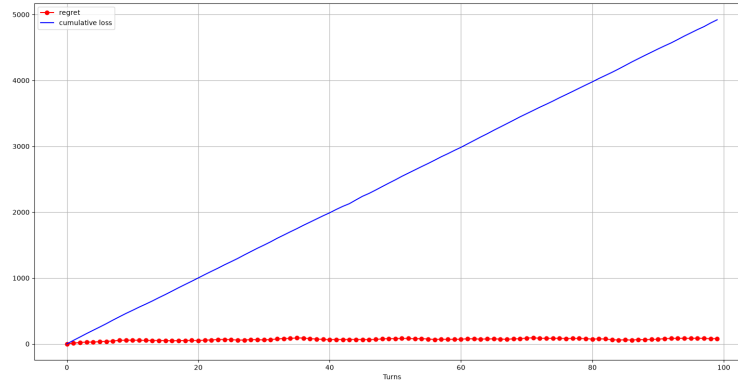


Fig. 8. Adaptive Hedge average regret=81.7

6 Conclusion

The analysis conducted shows that adaptive hedge performs significantly better in most cases especially if the tuning of the learning rate is possible using only limited information. The only situation where it doesn't shine is when the data aren't significant enough to infer a strategy (in our case low-gap).

Still in those worst cases it performs as bad as fixed hedge, as could be predicted given the properties of previously discussed.

References

1. Tim van Erven, Peter Grünwald, Wouter M. Koolen, Steven de Rooij (2011) *Adaptive Hedge*.

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